

# KAVINRAJ S

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## Education

### Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science — CGPA: 9.0

June 2023 – May 2027

Minor: Internet of Things

## Relevant Coursework

- Data Structures & Algorithms
- Machine Learning
- Computer Vision
- Natural Language Processing
- Database Systems
- Internet of Things

## Experience

### XCMG

Software Engineering Intern

May 2024 – July 2024

Tamil Nadu, India

- Engineered a real-time consumables tracking dashboard in Python that reduced reporting time by 75% and eliminated 12+ hours/week of manual data entry across 3 departments.
- Identified and resolved 5 critical workflow bottlenecks through cross-functional analysis, streamlining data pipelines and improving inventory accuracy by 40%.

### IEEE

Machine Learning Research Intern

Dec 2024 – Mar 2025

Remote

- Authored comprehensive Federated Learning case study analyzing privacy-utility tradeoffs across 8 benchmark datasets, published in IEEE research repository.
- Built automated FL simulation pipeline reducing experiment setup from 4 hours to 15 minutes, enabling 3x faster iteration on distributed model architectures.

## Projects

### Fraud Detection & Anomaly Analysis | Python, scikit-learn, TensorFlow, SMOTE, Streamlit

2025

- Engineered production-grade fraud detection system processing 284K+ transactions with 0.807 AUPRC, handling extreme class imbalance (0.17% fraud rate) through SMOTE oversampling and stratified validation.
- Optimized Random Forest classifier with automated threshold tuning, achieving 79.6% F1-score while reducing false positives by 23% compared to default configuration—saving estimated \$50K+ in manual review costs.
- Deployed interactive Streamlit dashboard with real-time prediction API, enabling non-technical stakeholders to test models on 1000+ sample transactions with sub-second latency.

### Privacy-Preserving Federated Learning System | Python, PyTorch, FLwr

2024

- Architected distributed federated learning system training models across 10+ simulated clients, achieving 96.8% accuracy (within 2% of centralized baseline) while maintaining local data privacy for sensitive healthcare/financial datasets.
- Implemented FedAvg aggregation strategy processing 50K+ distributed training samples, demonstrating scalability of privacy-preserving ML for production environments with strict data governance requirements.
- Benchmarked communication efficiency achieving 40% bandwidth reduction through gradient compression, addressing key deployment constraint for edge device scenarios.

### NEAT-Drive — Self-Driving Car Simulation | Python, NEAT, Reinforcement Learning

2025

- Implemented NEAT (NeuroEvolution of Augmenting Topologies) algorithm from scratch to evolve autonomous driving controllers, achieving 95%+ lane-keeping accuracy over 100+ track simulations without supervised labels.
- Designed multi-objective fitness function balancing safety, speed, and stability; optimized mutation/crossover rates achieving 35% faster convergence (50 vs 77 generations) to stable driving policies.
- Reduced collision rate from 45% (random policy) to 8% through evolutionary topology optimization, demonstrating effectiveness of genetic algorithms for complex control tasks.

### AI-Powered Unit Testing Generator | Python, TypeScript, LLMs, VS Code API

2025

- Built VS Code extension to auto-generate unit tests, achieving 85%+ code coverage and reducing manual test writing time by 60% across 20+ Python/TypeScript projects.
- Engineered context-aware code analysis system identifying edge cases and generating 200+ test cases per session with 92% validity rate, improving code quality for development teams.
- Integrated intelligent report generation pipeline creating coverage analysis documentation, saving developers 5+ hours/week in QA workflows.

- Built CUTS-based deep learning model for breast ultrasound tumor segmentation achieving 92%+ Dice coefficient on 500+ annotated images, supporting early cancer detection workflows.
- Developed automated preprocessing pipeline handling mask normalization and label conversion across 3 annotation schemas, reducing dataset preparation time by 80%.
- Benchmarked 4 model variants across malignant/benign classes, establishing robust baseline for clinical deployment with 89% sensitivity on malignant cases.

## Technical Skills

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**Languages:** Python, C, C++, Java, Go, JavaScript, SQL

**ML & AI:** PyTorch, TensorFlow, scikit-learn, FLwr, OpenCV, LangGraph, SMOTE, imbalanced-learn

**Developer Tools:** Git, Vim, VS Code, GCP, Streamlit

**Operating Systems:** Linux (Arch), Windows